

Sorting

Sorting is a process of arranging the elements in a particular order.

Sorting is mainly used to access the data more efficiently. For example, in the telephone directory the names will be sorted in alphabetical order, such that the process of searching is simplified.

Consider another example of books arranged in a specific order in a library so that books can be retrieved easily.

Basic Terminology: A file is a sequence of n items $r[0], r[1], \dots, r[n-1]$. Each item in a file is called a record.

There is a key associated with each record. This key is used to sort the records in a specific order (either ascending or descending).

The file is said to be sorted on the key $K[i]$, associated with i^{th} record precedes the key $K[j]$, associated with j^{th} record in the file based on some ordering on keys.

There are two types of sorting techniques.

1. Internal Sorting
2. External Sorting

If all the records to be sorted are present in the main memory, then such type of sorting is called internal sorting.

If all the records to be sorted are present in auxiliary storage, then such type of sorting is called external sorting.

Efficiency consideration:

There are various methods available for sorting. But we must be very careful while selecting a particular sorting technique.

There are three more important considerations one should look at during the selection of a sorting technique.

They are,

1. Coding Time: The amount of time required to write full length sorting program.
2. Execution Time: The amount of machine time necessary for executing the program.
3. Space: The amount of memory space necessary for the program.